SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: De**ign Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:20
Duration : 1 Hour

Scenario Based

Code of Conduct:

I B.Nynika(192311075) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B.Nynika

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1. Given a DFA used for validating a specific PIN pattern bbbc, bca ,c, ca, caaaaaa minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

Given Strings

- **bbbc**
- **bca**
- **c**
- **ca**
- **caaaaaa**

These are the strings accepted by the DFA.

Since the complete DFA transition diagram is **not provided**, the exact minimized DFA **cannot be constructed uniquely**. DFA minimization requires the original transition table or state diagram.

General Steps of DFA Minimization

1. Remove unreachable states.
2. Divide states into:
    - Final states
    - Non-final states
3. Check equivalent states.
4. Merge equivalent states.
5. Draw the minimized DFA.

Advantages of DFA Minimization

State reduction provides several benefits:

- Reduces memory usage.
- Improves execution speed.
- Requires fewer flip-flops in hardware.
- Consumes less power.
- Simplifies circuit design.
- Increases efficiency in embedded systems.

Applications

- ATM PIN verification
- Smart cards
- Password validation
- Embedded security systems
- Digital locks

2. Demonstrate how a given finite automaton can be converted into a regular expression

Automaton

States

- q_1 (Start & Final)
- q_2
- q_3
- q_4 (Trap)

Transitions

- $q_1 \rightarrow q_2$ on **0**
- $q_2 \rightarrow q_1$ on **1**
- $q_1 \rightarrow q_3$ on **1**
- $q_3 \rightarrow q_1$ on **0**
- $q_2 \rightarrow q_4$ on **0**
- $q_3 \rightarrow q_4$ on **1**
- q_4 loops on **0,1**

Observation

The automaton accepts

- **01**
- **10**

and any repetition of these pairs.

Thus the regular expression is

$$\boxed{(01 + 10)^*}$$

Examples Accepted

ε

01

10

0110

1001

01100110

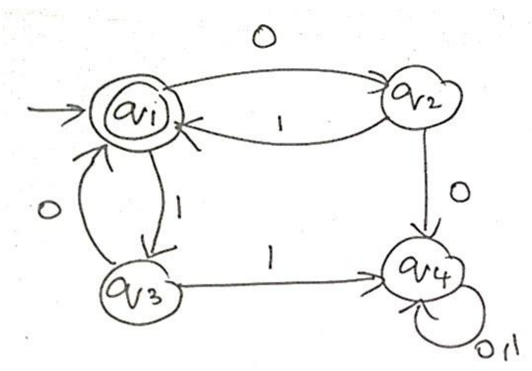
Examples Rejected

00

11

001

111



3. Consider the following grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

C Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

Given

S → aSc | T | U | ε

T → bTc | ε

U → aUb | ε

(A) Language Defined

Rule 1

S → aSc

Generates

$$a^n S c^n$$

Rule 2

T → bTc | ε

Generates

$$b^n c^n$$

Rule 3

U → aUb | ε

Generates

$$a^n b^n$$

Hence the grammar generates strings like

ε

ab

aabb

aaabbb

bc

bbcc

bbbccc

abc

aabbcc

aaabbbccc

(B) Derivation for aaabbc

Leftmost derivation

S

⇒ aSc

⇒ aaScc

⇒ aaUcc

⇒ aaaUbcc

⇒ aaabbcc

⇒ aaabbcc

Derivation Tree

```
     S
   / | \
  a  S  c
    / | \
    a U  c
     /|\
     a U b
       |
       ε
```

(C) Is the Grammar Ambiguous?

Answer

No

Reason:

Every string has only one valid derivation.

Therefore,

The grammar is not ambiguous.

4.I)Show that if L1 and L2 are regular then (L1+l2) also regular

Ii)Is the language { a^p| p is prime} regular? Justify

Suppose

- L_1 is accepted by DFA M_1
- L_2 is accepted by DFA M_2

Construct an NFA having ε-transitions from a new start state to M_1 and M_2.

This NFA accepts

$$L_1 \cup L_2$$

Since every NFA has an equivalent DFA,

$$L_1 + L_2$$

is regular.

Hence proved.

4(ii) Is

$$L = \{a^p \mid p \text{ is prime}\}$$

regular?

Answer

No

Proof using Pumping Lemma

Assume L is regular.

Let pumping length be **m**.

Choose

$$w = a^p$$

where **p** is a prime greater than **m**.

Split

$$w = xyz$$

such that

- $| xy | \leq m$
- $| y | > 0$

Pump y twice.

The new string length becomes

$$p + | y |$$

which is generally **not prime**.

Therefore,

the pumped string is **not in L**.

Contradiction.

Hence,

$$\boxed{L = \{a^p \mid p \text{ is prime}\} \text{ is not regular.}}$$

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand